
«utility functions»
EventSystemUtilities
+get_dispatcher() : EventDispatcher
+dispatchEvent(_type: EventType, data: Any) : void
+subscribeEvent(_type: EventType, callback: Callable) : void
+unsubscribeEvent(_type: EventType, callback: Callable) : void
delegates to singleton
EventDispatcher
- _instance: Optional[EventDispatcher]
- _lock: threading.Lock
- _subscribers: Dict[EventType, Set[Callable]]
- _last_events: Dict[EventType, Any]
- _subscribers_lock: threading.Lock
+get_instance() : EventDispatcher
+ __init__() : void
+subscribeEvent(_type: EventType, callback: Callable) : void
+unsubscribeEvent(_type: EventType, callback: Callable) : void
+dispatchEvent(_type: EventType, data: Any) : void
singleton instance
«utility»
Enum
+auto() : Int
inherits
«enumeration»
EventType
+INPUT: auto()
+STATE_CHANGE: auto()
+RECORDING_STARTED: auto()
+RECORDING_STOPPED: auto()
+JOINT_MODE_CHANGED: auto()
+COLLECTOR_CHANGED: auto()
+TRACKED_KEYS_CHANGED: auto()
uses for thread safety
manages subscriber sets
stores subscribers and last events
manages callback functions
dispatches data following patterns
notifies different subscriber types
implements
categorizes
categorizes
categorizes
«utility»
threading
+Lock
+RLock
«typing»
Set
+add(item: Any) : void
+remove(item: Any) : void
+__contains__(item: Any) : bool
«typing»
Dict
+__getitem__(key: Any) : Any
+__setitem__(key: Any, value: Any) : void
+__contains__(key: Any) : bool
«typing»
Callable
+__call__(*args: Any, **kwargs: Any) : Any
«data patterns»
EventDataPatterns
+InputEventData: "InputData object with collector info"
+StateChangeData: "Old and new state values"
+CollectorChangeData: "Collector type and capabilities"
+TrackedKeysData: "List of currently tracked keys"
+RecordingStateData: "Recording status and metadata"
«usage patterns»
SubscriberPatterns
+UISubscribers: "Update visual components"
+APISubscribers: "Broadcast to web clients"
+LoggingSubscribers: "Record events to files"
+ServiceSubscribers: "Update internal state"
«safety mechanisms»
ThreadSafetyMechanisms
+Class Lock: "Singleton instance creation"
+Subscribers Lock: "Subscription management"
+Exception Isolation: "Prevent callback errors from affecting others"
+Last Event Storage: "State synchronization for late subscribers"
«event category»
InputEvents
+INPUT: "Raw input events from collectors"
+TRACKED_KEYS_CHANGED: "Changes to tracked keys list"
«event category»
StateEvents
+STATE_CHANGE: "General application state changes"
+JOINT_MODE_CHANGED: "Edit mode toggle events"
+COLLECTOR_CHANGED: "Active collector switch events"
«event category»
RecordingEvents
+RECORDING_STARTED: "Recording session start"
+RECORDING_STOPPED: "Recording session end"